Sahaj Bhadja

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Summary

Software engineer with 2+ years building cloud-backed APIs, distributed data workflows, and multi-tenant systems across AWS, Azure, SQL, FastAPI, Flask, CI/CD, and fault-tolerant backend services.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and optimized a multi-tenant PostgreSQL schema for 200+ users and 10K+ structured records, using indexing, validation, and query tuning to reduce latency by 35% in production.
- Built ETL-style pipelines across FastAPI, SQL Server, Supabase, and cloud storage, processing 5K+ monthly jobs with reconciliation checks, error handling, and CI/CD-backed releases for reliable operations.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Implemented a production backend on AWS serving 150+ instructors and 10K+ daily grading requests, with CI/CD workflows, secure file ingestion, and stable API performance during peak load.
- Designed a concurrent batch-processing pipeline using multithreading and REST APIs to process 1K+ files per batch, maintaining fault-tolerant execution across distributed grading workflows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a GPT-based Decision Transformer for offline reinforcement learning, improving long-horizon trajectory planning by 18% while developing data pipelines processing 100K+ logs across large simulation workloads.
- Architected a spatial-temporal encoder integrating map geometry with ego-agent context through multi-head attention, improving closed-loop planning stability by 15% across complex multi-agent scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform data-sharing software with microservices and automated validation workflows, using Jenkins, PyTest, and Mockito to strengthen release reliability across Windows, Linux, and macOS.
- Troubleshot API integrations with AWS Glue and S3, improving response times by 30% while leading ETL workflow changes that boosted pipeline efficiency 68% and reduced technical debt 20%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based resource management platform with REST APIs, WebSockets, and microservices, supporting six workflows and 2M+ synchronized records while reducing code-review turnaround by 25% across releases.
- Automated deployments with Jenkins CI/CD and Docker packaging, achieving 95%+ successful build validation and cutting deployment time by 50% for cloud-backed application updates.

Skills

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, Bash
Systems: Distributed systems, microservices, multithreading, cloud architecture, AWS, Azure, GCP, S3, AWS Glue, Azure SQL, Blob Storage, Docker, Kubernetes, ECS/Fargate, FastAPI, Flask, REST APIs
Data & Delivery: PostgreSQL, SQL Server, MySQL, MongoDB, ETL pipelines, indexing, query tuning, data validation, reconciliation checks, CI/CD, Jenkins, GitHub Actions, PyTest, debugging, fault isolation, monitoring, system design, algorithms, Agile/Scrum, Git

Projects

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Engineered a cloud ingestion and indexing pipeline processing 50K+ records, implementing incremental updates, metadata persistence, and tuning to reduce retrieval latency by 40% for document workloads.

FinNexus Live | *Next.js, TypeScript, Kubernetes, Plaid*
- Built a scalable fintech platform supporting 50,000+ users and 1M+ transactions, combining real-time updates, Kubernetes deployments, and secure workflows to maintain reliable high-volume distributed behavior.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed and deployed a containerized RESTful data processing API on AWS using CodePipeline and ECS/Fargate, with automated CI/CD releases, validation checks, and reliable recurring processing workflows.